

Anthony Minh Nguyen

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Education

University of California, San Diego

La Jolla, CA

B.S. in Electrical Engineering, **GPA: 3.988/4.000**

Sep 2023 - Jun 2026 (Expected)

Depth: Machine Learning and Controls

Relevant Coursework: Machine Learning; Generative Models; Control Theory; Optimization; Probability Theory; Random Processes; Digital Signal Processing; Linear Systems; Electromagnetism; Digital Design

Research Interests

Distributed systems; distributed optimization; distributed control; multi-agent systems; swarm systems; decentralized systems; reinforcement learning; partially observable Markov decision processes (POMDPs); planning under uncertainty; state estimation; Bayesian inference and filtering; optimal and stochastic control; robotics; generative models; machine learning; optimization

Research Experience

Existential Robotics Laboratory (UC San Diego)

La Jolla, CA

Undergraduate Student Researcher

Jan 2025 - Present

- Worked under Professor Nikolay A. Atanasov
- Developed Monte Carlo Tree Search (MCTS) algorithms for optimal control of partially observable Markov decision processes (POMDPs), with applications to sensor scheduling
- Discovered, formalized, and proved relationships among multiple particle flow particle filtering (PFPF) formulations; derived methods to mitigate numerical instability
- Implemented PFPF algorithms in Python for linear and nonlinear observation models using both analytical and numerical approaches
- Solved continuous-time particle flow dynamics using ODE solvers (SciPy, Diffrax) and GPU-accelerated computation (JAX)
- Completed independent graduate-level study in decision theory, Bayesian inference, Bayesian filtering, Kalman filtering, particle filtering, MDPs, POMDPs, optimal control, and reinforcement learning

Cognicom, Inc.

San Diego, CA

Visiting Researcher

Apr 2025 - Present

- Investigated diffusion-based generative models for realistic RF signal synthesis to support RF machine learning and cognitive radio systems.
- Conducted literature reviews on generative modeling and robotics to support integration of AI technologies within defense-oriented systems.

IEM OPALS (UC San Diego)

La Jolla, CA

Research Intern

Jun 2022 - Aug 2022

- Worked under Dr. Linda Shi (Co-director of the UCSD Biophotonics Technology Center)
- Troubleshooted, assembled, and contributed to LabVIEW programming of cell microsurgery robotic systems based on UCSD Biophotonics Lab publications
- Collaborated with a graduate student to validate experimental results for a project investigating DNA repair mechanisms in cells treated with red light

Projects

FadeGen: Diffusion Model for Generating Realistic Radio-Frequency Signal Data

Probabilistic Graphical Models Final Project, Research for Cognicom

2025

- Transformer-based denoising diffusion probabilistic model (DDPM) for generating radio-frequency (RF) signals corrupted with channel fading
- Implemented using PyTorch

Industry Experience	L3Harris Technologies, Inc. Electrical Engineering Intern Carlsbad, CA Jun 2025 - Sep 2025 – Migrated a communication system design from Altera to Xilinx platforms; updated RTL for new hardware constraints – Re-implemented transmission flagging logic for the revised architecture – Presented internal seminars on recent advances in generative models and robotics
	Cognicom, Inc. Electrical Engineering Intern San Diego, CA Jan 2023 - Sep 2023 – Designed and implemented serial communication protocols and a precision timer circuit in Verilog – Developed a bare-metal C program for Xilinx Zynq to control an interface for a tactical radio system
Technical Skills	Languages: Python, C, Java, Verilog, VHDL Tools: NumPy, PyTorch, JAX, TensorFlow, MATLAB, QuestaSim, Active-HDL, Xilinx Vivado, L ^A T _E X/Overleaf, Ubuntu/RedHat Linux
Mathematical & Theoretical Skills	• Advanced Calculus - multivariable, vector • Differential Equations - ordinary differential equations, continuous-time dynamical systems • Linear Algebra - least squares, matrix calculus, numerical, analytical • Optimization - gradient descent, Gauss-Newton, Levenberg-Marquardt, convexity, unconstrained • Probability and Random Processes - Decision-Theoretic Models (MDPs, POMDPs), Markov Chains, Monte Carlo methods • State Estimation - Bayesian inference, Bayesian filtering, Kalman filtering (Extended, Unscented), Particle (flow) filtering
Communication & Writing	• Ability to formulate and articulate research problems with mathematical and theoretical clarity • Experienced in academic and technical writing, including the preparation of research reports, proofs, and manuscript-style documents • Proficient in presenting complex theoretical and experimental results in a clear, structured, and rigorous manner • Familiar with L^AT_EX-based academic writing workflows for figures, equations, and citations
Honors & Awards	• UC San Diego Provost Honors Fall 2023 - Present • Salute to Excellence Award May 2023 (Top 1% of San Diego Unified School District Class of 2023) • University of Rochester George Eastman Young Leaders Award May 2022